

Weekly Report

What I Worked On

Profiling

ABINIT

- **Path:** abinit/v2/fullPEAK-scaling-07102026-152428/timing_summary.csv

Job Name	Config	Nodes	Tasks	PEAK?	Elapsed (s)	Exit Code	Job ID
abinit_test_n1_nopeak	n1_nopeak	1	1	No	119	0	3294914
abinit_test_n24_nopeak	n24_nopeak	1	24	No	10	0	3294915
abinit_test_n48_nopeak	n48_nopeak	1	48	No	8	0	3294916
abinit_test_N2_nopeak	N2_nopeak	2	96	No	6	0	3294917
abinit_test_N8_nopeak	N8_nopeak	8	384	No	6	0	3294918
abinit_test_N16_nopeak	N16_nopeak	16	768	No	9	0	3294919
abinit_test_n1_peak	n1_peak	1	1	Yes	126	0	3294920

Function	Calls	Total Time (s)	Avg Time / Call (ms)	PEAK Overhead (s)
zgemm_	207,990	4.7613	0.0229	0.4258
ddot_	216,544	0.6383	0.0029	0.4433
zcopy_	131,022	0.6081	0.0046	0.2682
dznrm2_	116,280	0.5958	0.0051	0.2380
zdotc_	51,588	0.1846	0.0036	0.1056

Takeaway

- ABINIT effectively scaled from 1 to 96 tasks but performs worse when scaled further on this test case.
- As I understand it, the decrease in speed up is most likely due to inter-node communication overhead.

Quantum Espresso

- Path: qe/fftw_test-scaling-07142026-092819/timing_summary.csv

Job Name	Config	Nodes	Tasks	PEAK?	Elapsed (s)	Exit Code	Job ID
qe_tc_n1_	n1_	1	1	No	636	0	3306688
nopeak	nopeak						
qe_tc_n24_	n24_	1	24	No	63	0	3306689
nopeak	nopeak						
qe_tc_n48_	n48_	1	48	No	34	0	3306690
nopeak	nopeak						
qe_tc_N2_	N2_	2	96	No	28	0	3306691
nopeak	nopeak						
qe_tc_N8_	N8_	8	384	No	20	0	3306692
nopeak	nopeak						
qe_tc_N16_	N16_	16	768	No	27	0	3306693
nopeak	nopeak						
qe_tc_n1_peak	n1_	1	1	Yes	638	0	3306694
	peak						

Function	Calls	Total Time (s)	Avg Time / Call (ms)	PEAK Overhead (s)
zgemm_	144,849	166.6082	1.1502	0.2227
zhgevx_	5,561	8.3896	1.5087	0.0085
zgemv_	1,184	3.7863	3.1979	0.0018
ddot_	57,984	0.3923	0.0068	0.0891
dcopy_	1,926	0.3207	0.1665	0.0030

Takeaway

- Similiar story to ABINIT

DFTB+

- Path: dftb/new_gen-scaling-07132026-145728/timing_summary.csv

Job Name	Config	Nodes	Tasks	PEAK?	Elapsed (s)	Exit Code	Job ID
dftb_	n1_	1	1	No	120	0	3303873
spinlock_n1_	nopeak						
nopeak							

Job Name	Config	Nodes	Tasks	PEAK?	Elapsed (s)	Exit Code	Job ID
dftb_ spinlock_n24_ nopeak	n24_ nopeak	1	24	No	144	0	3303874
dftb_ spinlock_n48_ nopeak	n48_ nopeak	1	48	No	220	0	3303875
dftb_ spinlock_n1_ peak	n1_ peak	1	1	Yes	121	0	3303879
dftb_ spinlock_N2_ nopeak	N2_ nopeak	2	96	No	2	1	3303876
dftb_ spinlock_N8_ nopeak	N8_ nopeak	8	384	No	3	1	3303877
dftb_ spinlock_N16_ nopeak	N16_ nopeak	16	768	No	3	1	3303878

Function	Calls	Total Time (s)	Avg Time / Call (s)	PEAK Overhead (s)
dsyev_	4	94.1030	23.5258	0.0000
dgemm_	442	10.5098	0.0238	0.0007
dsymm_	4	10.0370	2.5092	0.0000
dlarrv_	4	0.1037	0.0259	0.0000
dcopy_	34,760	0.0280	0.0008	0.0541

Takeaway

- DFTB+ seemingly performed worse as it scaled on this test case, even crashing on multi-node configurations.
- Im going to look into this one more as it doesn't make sense to me.

LAMMPS

- **Path:** lammps/5-scaling-07132026-225745/timing_summary.csv

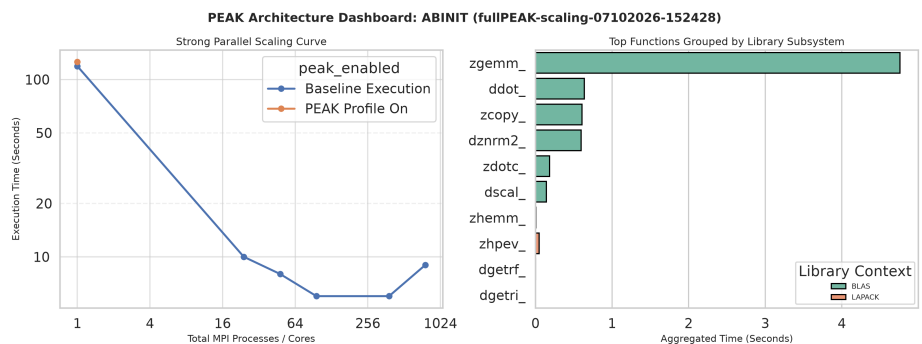


Figure 1: plot

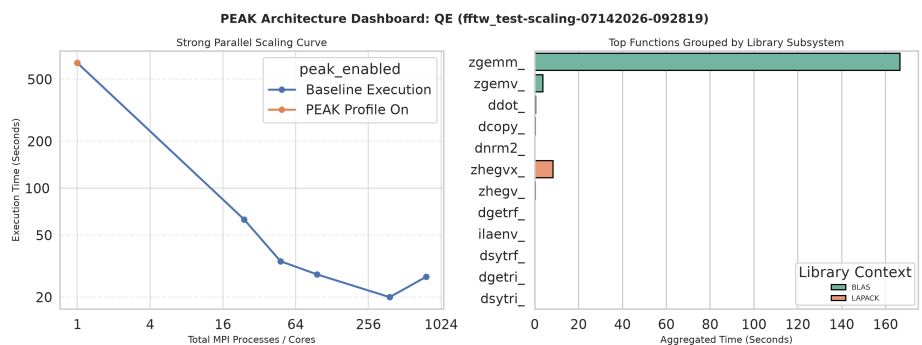


Figure 2: plot

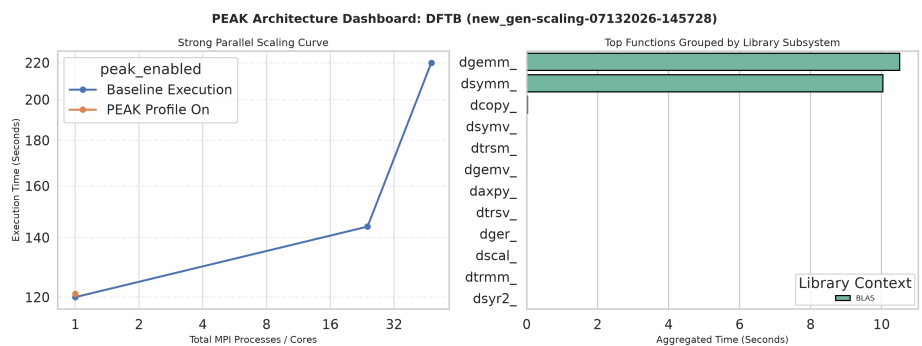


Figure 3: plot

Job Name	Config	Nodes	Tasks	PEAK?	Elapsed (s)	Exit Code	Job ID
lammps_rhodo_n1_nopeak	n1_nopeak	1	1	No	32	0	3305579
lammps_rhodo_n24_nopeak	n24_nopeak	1	24	No	4	0	3305580
lammps_rhodo_n48_nopeak	n48_nopeak	1	48	No	4	0	3305581
lammps_rhodo_N2_nopeak	N2_nopeak	2	96	No	4	0	3305582
lammps_rhodo_N8_nopeak	N8_nopeak	8	384	No	4	0	3305583
lammps_rhodo_N16_nopeak	N16_nopeak	16	768	No	8	255	3305584
lammps_rhodo_n1_peak	n1_peak	1	1	Yes	32	0	3305585

LAMMPS did not yield any PEAK hits on both module and build script versions.

Takeaway

- LAMMPS scaled great, but seemingly limited itself after an upper limit.
- As I understand it, this limit is based on the input and more complex input files will scale better.

GROMACS

- **Path:** gromacs/2-scaling-07102026-221528/timing_summary.csv

Job Name	Config	Nodes	Tasks	PEAK?	Elapsed (s)	Exit Code	Job ID
gromacs_benchMEM_n24_nopeak	n24_nopeak	1	24	No	213	0	3295859
gromacs_benchMEM_n48_nopeak	n48_nopeak	1	48	No	125	0	3295860
gromacs_benchMEM_n48_peak	n48_peak	1	48	Yes	126	0	3295864
gromacs_benchMEM_N2_nopeak	N2_nopeak	2	96	No	75	0	3295861

Job Name	Config	Nodes	Tasks	PEAK?	Elapsed (s)	Exit Code	Job ID
gromacs_ benchMEM_N8_ nopeak	N8_ nopeak	8	384	No	37	0	3295862
gromacs_ benchMEM_N16_ nopeak	N16_ nopeak	16	768	No	36	0	3295863

GROMACS did not yield any PEAK hits on both module and build script versions.

Takeaway

- GROMACS seemingly takes great advantage of scaling.

Created another script to parse research data and create graphs

What I Learned

- Creating input files
- Working with QE, DFTB+, LAMMPS, and GROMACS
- Interpreting profiling data

Questions / Concerns

- For cases where PEAK does not get any hits for BLAS, LAPACK, and FFTW. Is there anything else you would want me to do or try?
 - From what I've looked into, it may be due to the way these programs were compiled as for why I'm getting no hits.
- For DFTB+, I am using the spinlock case for testing. Is there a different test case I could use?

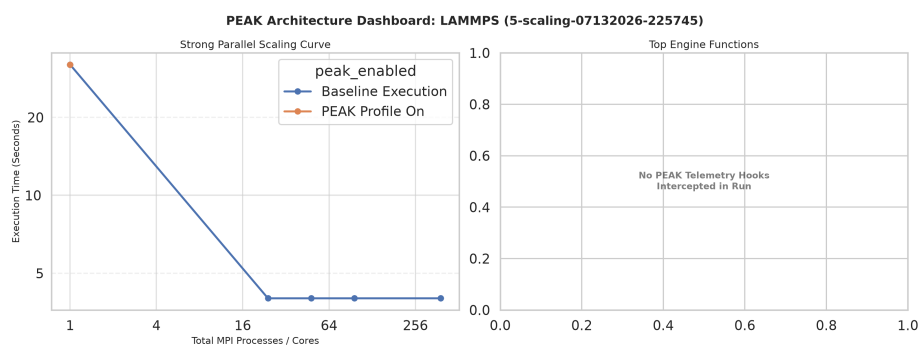


Figure 4: plot

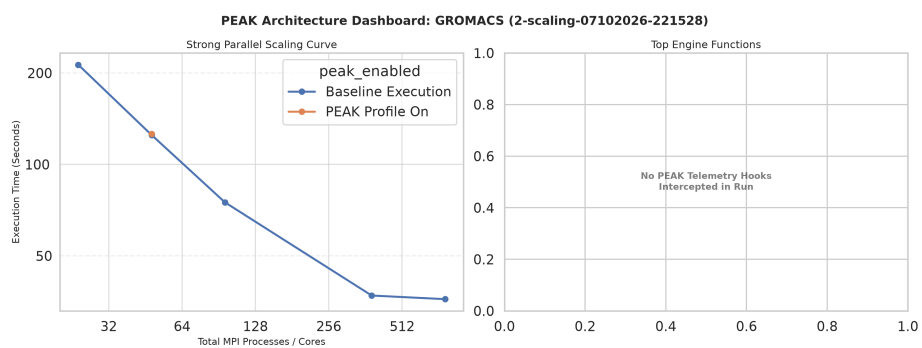


Figure 5: plot